

Biodegradation of hydrocarbons by *Purpureocillium lilacinum* and *Penicillium chrysogenum* from heavy oil sludge and their potential for bioremediation of contaminated soils

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ABSTRACT

Heavy oil sludge as a highly hazardous solid waste poses potential pollution risks to the ecological environment. Microbial degradation of petroleum hydrocarbons is an emerging strategy for controlling the environmental pollution by oil sludge. Two hydrocarbon-degrading fungi, designated HF and HQ, were isolated from heavy oil sludge collected at Tahe Oilfield, Xinjiang, Northwest China. Based on morphological analysis and gene sequencing, HF and HQ were identified as *Purpureocillium lilacinum* and *Penicillium chrysogenum*, respectively. When supplemented with a small amount of additional carbon (soluble starch) and nitrogen (yeast extract) sources, HF and HQ were able to biodegrade hydrocarbons in oil sludge. Following fungal treatment (30 days), the relative abundances of various hydrocarbon components changed in distinct patterns based on gas chromatography-mass spectrometry. The mean degradation rate of HF for cycloalkanes, *n*-alkanes, aromatics, and *iso*-alkanes was 100.0%, 21.2%, 15.1%, and 14.5%, respectively. HQ showed contrasting effects on hydrocarbon composition, as indicated by the decrease of *n*-alkanes (1.3%) and the increase of cycloalkanes, aromatics, and *iso*-alkanes (82.6%, 10.2%, and 11.9%, respectively). Approximately 15.3% and 7.6% of total petroleum hydrocarbons in modified oil sludge were removed by HF and HQ, respectively. The results provide empirical evidence for application of *P. lilacinum* HF and *P. chrysogenum* HQ as candidate hydrocarbon degraders in bioremediation of oil sludge-contaminated soils.

1. Introduction

Heavy crude oil, a key component of global energy supply, has attracted increasing attention in recent decades (Dong et al., 2019). In the wake of heavy oil exploitation worldwide, massive amounts of oil sludge are inevitably produced and released into the soil environment (Ramirez et al., 2019). Exposure to oil sludge destroys soil structure and deteriorates soil quality, in turn hindering plant growth. Additionally, the contaminants in oil sludge can migrate from the soil into ground water, posing a threat to drinking water safety and human health (Su et al., 2021). Owing to its toxic, mutagenic, and carcinogenic properties, oil sludge has been classified as a priority environmental pollutant by the US Environmental Protection Agency, with strict control over its release into the environment (Zhao et al., 2019).

Oil sludge is mainly composed of various petroleum hydrocarbons, water, solid particles, and heavy metals. This complex mixture persists

in the environment because of its high hydrocarbon content, strong viscosity, difficult dehydration, and serious emulsification (Ramirez et al., 2019; Su et al., 2021). Myriad studies have been conducted to treat oil sludge using chemical, physical, and biological techniques (Zhang et al., 2022). Physical and chemical techniques yield favorable results in terms of hydrocarbon removal and transformation. Although both the techniques remove a substantial amount of hydrocarbons from oil sludge, the ultimate and complete hydrocarbon degradation is mainly accomplished by soil microorganisms (Sharuddin et al., 2021). This has led to the rapid development and wide application of biological techniques, particularly microbiological methods for oil sludge treatment (Chand et al., 2022).

Biologically-based techniques take advantage of microbial metabolic activity to clean up organic contaminants in soil under controlled conditions. The effects of hazardous contaminants are attenuated either by transforming them to an innocuous state or by reducing their

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concentrations to levels below the limits established by regulatory authorities (Yang et al., 2016). Microorganisms act as drivers of soil nutrient cycling and some taxa maintain the health of the soil by utilizing organic contaminants as carbon and energy sources (Haque et al., 2022). Microbiological remediation of organic contaminants via natural attenuation, biostimulation, and bioaugmentation is advantageous in terms of high economic applicability, environmental friendliness, and documented successful applications (Nwankwegu et al., 2022). While natural attenuation with slow biodegradation processes can rarely be used independently, bioaugmentation is more desirable for highly effective degradation (Saha et al., 2019; Essabri et al., 2019; Péquin et al., 2022). Compared with other microbiological methods, bioaugmentation enables greater degradation of hydrocarbons and their homologs (Bento et al., 2005; Nwankwegu et al., 2022). A pilot-scale study of bioaugmentation demonstrated that 76.0% of crude oil in soil was degraded by an exogenously seeded bacterial consortium, allowing crop plants to germinate and grow (Deng et al., 2020).

To remediate oil sludge-contaminated soils, the core of bioaugmentation is the screening of extraordinarily efficient hydrocarbon-degrading microorganisms from contaminated environmental matrices and plants originally grown in contaminated sites (Baoune et al., 2019; Ehiosun et al., 2022). Generally, bacteria are considered to play a major role in biodegradation and mineralization of hydrocarbons due to their versatility. However, oil sludge contains abundant petroleum hydrocarbons, including long-chain *n*-alkanes, polyaromatic hydrocarbons, resins, and asphaltenes, which are recalcitrant to microbial degradation (Chand et al., 2022). Consequently, biodegradation of recalcitrant hydrocarbons in oil sludge necessitates sophisticated biomolecular machinery of biosurfactants and participation of synergic enzymes (Rehman et al., 2021).

Different cassettes of unspecific enzymes involved in hydrocarbon degradation can be synthesized by a vast diversity of fungi. In addition to intracellular enzymes (such as cytochrome P450 monooxygenases, nitroreductases, and transferases), extracellular enzymes (laccases, hydrolases, and peroxidases) from fungi are capable of degrading long-chain petroleum hydrocarbons and complex polycyclic aromatics (Gnanasekaran et al., 2019; Medaura et al., 2021). The mycelial structure of fungi with an invasive nature and large surface area gives them another advantage over bacteria for bioremediation of contaminated soils (Benguenab and Chibani, 2021). For example, Pourfakhraei et al. (2018) demonstrated that incubation of heavy oil with *Daedaleopsis* sp. EF5 resulted in increased contents of $<C_{24}$ hydrocarbons and decreased contents of $>C_{24}$ compounds. *Aspergillus tubigenis* FGS-4 with laccase activity was also reported as an efficient hydrocarbon degrader, which completely degraded primary hydrocarbons in crude oil to produce short-chain compounds, such as ketones and alcohols (Gnanasekaran et al., 2019). The use of hydrocarbon-degrading fungi thus provides an effective solution for remediating the soil in oil sludge-contaminated sites. In some circumstances, fungi-based techniques work better than traditional bioremediation techniques involving bacteria (Bovio et al., 2017).

The filamentous fungi with hydrocarbon degradation activity are exemplified by *Purpureocillium* spp. and *Penicillium* spp. (Al-Hawash et al., 2018; Mikolasch et al., 2019). In particular, *Penicillium* spp. that thrive in contaminated sites have received much attention because of their high bioremediation potential (Bovio et al., 2017; Al-Hawash et al., 2018). For example, *Penicillium citreonigrum* MUT 267 isolated from a Mediterranean marine site after an oil spill induced a conspicuous decrease of low-molecular-weight *n*-alkanes (C_{14} – C_{18}) and demonstrated a moderate ability to remove high-molecular-weight *n*-alkanes ($>C_{30}$) (Bovio et al., 2017). Additionally, two *Penicillium* sp. strains, RMA1 and RMA2, isolated from an Iraqi oil field performed effective biodegradation and emulsification of crude oil (Al-Hawash et al., 2018). In contrast with *Penicillium* spp., *Purpureocillium* spp. are rarely described as efficient degraders of crude oil hydrocarbons. *Purpureocillium* spp. are commonly found in the soil or rotting materials. They

have wide application in the biological control of plant-parasitic nematodes and insects (Gallego-Velázquez et al., 2014) and are considered emerging plant growth-promoting bioinoculants (Baron et al., 2020). Recently, Mikolasch et al. (2019) isolated a number of strains from oil- and hydrocarbon-polluted soils in Kazakhstan, and among them, *Purpureocillium lilacinus* SBUG-M 1730 showed degradation activity through transformation of tetradecane into dicarboxylic acid. Furthermore, Benguenab and Chibani (2021) isolated a *P. lilacinum* strain from a used engine oil-contaminated soil, which degraded 44.6% of crude oil.

Heavy oil sludge provides an excellent source of microorganisms with high potential for partial or complete degradation of recalcitrant hydrocarbons. In the present study, two hydrocarbon-degrading fungi classified as *P. lilacinum* and *P. chrysogenum* were isolated from an oil sludge sample. The bioremediation potential of the two indigenous fungi was systematically evaluated by hydrocarbon biodegradation and simulated bioremediation tests. Findings of this study implicate the fungi as two candidate strains for bioremediation of oil sludge-contaminated soils.

2. Materials and methods

2.1. Oil samples and culture media

The heavy oil sludge used in the study was collected from pipeline facilities in an oil well of the No. 2 Oil Production Plant, Tahe Oilfield, Xinjiang, China. The oil sludge sample had a pH of 7.79, salt content of 13.78 g L⁻¹, water content of 11.10%, solid phase content of 61.98%, and total petroleum hydrocarbon (TPH) content of 269.2 g kg⁻¹. Petroleum hydrocarbons were extracted from the oil sludge sample using the method of Othman et al. (2020), with minor modifications. The mean contents of saturates, aromatics, resins, and asphaltenes in the petroleum hydrocarbons were 10.73 ± 1.64%, 29.98 ± 2.37%, 1.92 ± 0.20%, and 57.83 ± 1.20%, respectively. Light crude oil was collected from Kelamayi Oilfield, Xinjiang, and its mean contents of saturates, aromatics, resins, and asphaltenes were 75.33 ± 3.53%, 21.15 ± 0.64%, 1.72 ± 0.46%, and 2.57 ± 0.25%, respectively.

A mineral basal medium was prepared as previously reported (Zhang et al., 2016), with minor modifications. The mineral solution containing 2.00 g urea, 0.25 g MgSO₄·7H₂O, 5.00 g NaCl, 2.00 g KH₂PO₄, and 3.00 g K₂HPO₄·3 H₂O was dispersed in 1000 mL of tap water. Then, the mineral basal medium was supplemented with 100 g L⁻¹ oil sludge to formulate an enrichment medium. Potato dextrose agar (PDA) containing 200 g potato, 20 g dextrose, and 10–12 g agar was dissolved in 1000 mL of tap water (Cheng and Xue, 2012). The screening medium was prepared by adding 60 g L⁻¹ oil sludge and 10 g L⁻¹ agarose to the mineral basal medium. The biodegradation medium comprised 20 g L⁻¹ light oil or hydrocarbons extracted from oil sludge sample in the mineral basal medium, with 2 g L⁻¹ soluble starch as an auxiliary carbon source and 2 g L⁻¹ yeast extract as an auxiliary nitrogen source (Zhang et al., 2016). All the media were adjusted to pH 7.0 and sterilized by autoclaving at 121 °C for 30 min before use.

2.2. Isolation, identification, and medium optimization of hydrocarbon-degrading fungi

A 10.00 g sample of oil sludge was added to a 250-mL Erlenmeyer flask containing 100 mL of mineral basal medium. The sample was incubated at 30 °C with shaking at 120 r min⁻¹ for 7 days. Then, 5 mL of the culture was transferred to fresh enrichment medium and incubated for 7 days. This procedure was repeated three times. The third enrichment culture was 10-fold diluted and 50 µL aliquots were spread on PDA plates. After incubation at 30 °C, fungal colonies with different colors and shapes were picked for isolation and purification on fresh PDA plates using the dilution plating technique. Purified isolates were preserved on PDA slants and screened with oil sludge as the sole carbon source. The growth of colonies was observed and recorded after

incubation at 30 °C. The colony diameter was measured to determine hydrocarbon-degrading ability. Two isolates with the largest colony diameter were selected for subsequent analyses.

Morphological identification of these fungi was performed by colony observation and electron microscopy. The shape, color, and texture of 4-day-old colonies on PDA plates were observed. Additionally, samples were prepared by obliquely inserting glass slides into PDA plates inoculated with purified fungi. Four days later, conidiation and conidial morphology on the glass slides were observed using an S-4800 field emission scanning electron microscope (Hitachi, Tokyo, Japan).

Total genomic DNA of the fungi was extracted by the CTAB method as previously reported (Yanto and Tachibana, 2014). The ribosomal RNA (rDNA) internal transcribed spacer (ITS) gene was amplified by PCR using the fungal universal primer pair, ITS1 (5'-TCCGTAGGTGAACCTGCGG-3') and ITS4 (5'-TCCTCCGCTTATTGATAT-3'). The PCR products were electrophoresed in an 8 g L⁻¹ agarose gel and sent to Beijing Qingke Biotechnology Co., Ltd. (Beijing, China) for sequencing. The sequences were compared to sequence data from the GenBank database of NCBI (<http://www.ncbi.nlm.nih.gov/>) using the Basic Local Alignment Search Tool (BLAST) to determine their approximate phylogenetic affiliations. A phylogenetic tree was constructed based on the neighbor-joining method using MEGA v7.0 software with 1000 bootstraps (Al-Hawash et al., 2018). The rDNA-ITS sequences of two selected strains have been deposited in the GenBank database under accession numbers OP 529825 and OP 529826, respectively.

To select the optimal carbon source for the fungi, the screening medium was supplemented with six different carbon sources (each 0.2%, w/v): mannitol, lactose, sucrose, glucose, soluble starch, and heavy oil. To select the optimum nitrogen source, the screening medium omitting nitrogen source was supplemented with the optimal carbon source (each 0.2%, w/v) and five different nitrogen sources (each 0.2%, w/v): urea, (NH₄)₂SO₄, NH₄Cl, yeast extract, and peptone. Each fungal isolate was inoculated onto the screening medium and incubated at 30 °C for 14 days (carbon source) or 7 days (nitrogen source). The growth diameter of colonies was recorded on a daily basis. Three replications were prepared for each nutrient source.

2.3. Hydrocarbon biodegradation test

One inoculating loop of fungal culture was transferred from PDA slant to a 250-mL Erlenmeyer flask containing 100 mL of the biodegradation medium. The cultures were incubated at 30 °C for 30 days. The biodegradation medium without inoculation was used as the control. The degradation rate of TPH was determined gravimetrically, and the composition of hydrocarbons (saturates, alkenes, aromatics) was analyzed by gas chromatography-mass spectrometry (GC-MS). Each treatment had three replications.

After biodegradation, the culture broth containing residual TPH was filtered into another Erlenmeyer flask with absorbent cotton. The TPH-covered absorbent cotton was placed in the original Erlenmeyer flask and soaked with 30 mL of *n*-hexane to dissolve TPH. After that, the *n*-hexane solution containing residual TPH was filtered again with fresh absorbent cotton. In this way, the residual TPH was eluted completely with *n*-hexane according to solubility parameter close principle, and little TPH would be retained in the absorbent cotton. Next, the original Erlenmeyer flask, first absorbent cotton, and second absorbent cotton on the funnel were washed several times with 50 mL volumes of *n*-hexane. Each *n*-hexane solution containing residual TPH was transferred to the same 250-mL Erlenmeyer flask and rotary evaporated at 40 °C to constant weight. The degradation rate of TPH (VR_m%) by fungi was calculated using Formula (1):

$$VR_m\% = \frac{M_{Ctrl} - M_t}{M_{Ctrl}} \times 100 \quad (1)$$

where M_{Ctrl} and M_t (g) represent the mass of TPH in the control and

fungal treatment, respectively.

The residual TPH samples (1.000 g each) were dissolved in 15 mL of *n*-hexane. The chromatographic peak area of gasifiable hydrocarbon components was obtained to indicate their relative abundance. The changes in the relative abundance of hydrocarbon components (VR_a%) corresponding to different retention times was calculated by Formula (2):

$$VR_a\% = \frac{A_{Ctrl} - A_t}{A_{Ctrl}} \times 100 \quad (2)$$

where A_{Ctrl} and A_t are the chromatographic peak area of individual hydrocarbon components in the control and fungal treatment, respectively.

GC analysis was conducted using the Pegasus 4D-C gas chromatograph (LECO, Saint Joseph, MI, USA) equipped with a DB-PETRO column (60 m × 0.25 mm × 0.25 μm). Column temperature was initiated at 60 °C for 1 min and ramped up at 2 °C min⁻¹ to 310 °C, and held for 5 min. Other conditions included injector temperature, 280 °C; injection mode: splitless injection; injection volume, 0.2 μL; carrier gas: helium; flow rate, 1.0 mL min⁻¹. MS conditions were ionization mode, electron ionization; ionization energy, 70 eV; ion source temperature, 230 °C; interface temperature, 280 °C; scanning mode: full scan, scanning range, 35–650 amu; scanning time, 0.45 s; and solvent delay, 4 min.

2.4. Simulated bioremediation test

A mixture of wheat bran (100 g) and mineral basal medium (100 mL) was sterilized in 600 -mL tissue culture vessels by autoclaving at 121 °C for 30 min. After cooling, one inoculating loop of each fungal culture was transferred from PDA slant to the bran and stirred uniformly. The inoculated bran was incubated at 30 °C for 2–3 days to prepare the solid inoculum. Considering its recalcitrant components and dense texture, the oil sludge was not suitable for direct bioremediation. It was added with an inorganic bulking agent (sand grain) to adjust the compositional structure, improve the ventilation environment, and facilitate aerobic degradation. An organic bulking agent (wheat bran) was also added to improve the water retention capacity of oil sludge, increase the contact area between hydrocarbons and microorganisms, and provoke co-metabolism with oil sludge. Oil sludge, wheat bran, and sand grain were thoroughly mixed at a ratio of 1:8:3 (w/w/w), with the water content adjusted to 30% using the mineral basal medium. The TPH content of the modified oil sludge was 26.92 g kg⁻¹.

The modified oil sludge was dispensed into three groups of 600-mL tissue culture vessels (130 g each). Group 1 was a blank control with no fungal inoculation after sterilization. Group 2 received a 1% (w/w) inoculum of strain HF. Group 3 received a 1% (w/w) inoculum of strain HQ. There were three replications for each treatment group. All vessels were incubated at 30 °C for 30 days. To extract the residual hydrocarbons, 50.00 g of air-dried oil sludge was accurately weighed into a 100-mL centrifuge tube and 30 mL of *n*-hexane was added. Extraction was carried out for 20 min using an ultrasonic cleaner. The extract was centrifuged at 5000 r min⁻¹ for 5 min. The supernatant was poured into a 50-mL beaker of known weight. The extraction procedure was repeated three times. The third extract was re-extracted with a dichloromethane solution, and the final extract was rotary evaporated at 40 °C to constant weight. The removal efficiency and composition of hydrocarbons in the oil sludge were analyzed as described in subsection 2.3 Hydrocarbon biodegradation test.

2.5. Statistical analysis

SAS v9.2 (SAS Institute Inc, Cary, NC, USA) was used for one-way analysis of variance followed by Duncan's multiple range test (*P* < 0.05) to analyze the differences between group means. Graphs were produced using Origin v2022 (OriginLab Corp., Northampton, MA,

USA).

3. Results

3.1. Screening and identification of hydrocarbon-degrading fungi

Two hydrocarbon-degrading fungi, designated HF and HQ, were obtained through enrichment culture of oil sludge from Tahe Oilfield, Xinjiang. Both fungi were able to grow with oil sludge as the sole carbon source (Fig. 1a). Their capability to degrade hydrocarbons in oil sludge was preliminarily demonstrated by large colony diameters, which reached an average of 50 mm (HF) and 58 mm (HQ) at 14 days.

The colonial morphology and cellular micromorphology of the two fungi were distinctly different (Fig. 2). Strain HF formed round colonies and light purple conidia. Its colony surface was flat with regular radial grooves, powdery texture, and no exudate. Strain HQ formed round colonies and blue-green conidia. The colonies were characterized by a flat surface with regular radial grooves, villous texture, and considerable yellow exudate. Scanning electron microscopy revealed that the conidiophores of strain HF were characterized by a swollen base, with verticillate conical conidial phialides arising from the tip. The conidiophores of strain HQ were characterized by three verticillate penicillus, bearing phialides with lanceolate conidia. Based on the morphological observations, the two isolates were preliminarily identified as *Purpureocillium* and *Penicillium*, respectively (Kong, 2007).

Furthermore, the fungal sequences obtained in this study were aligned against the NCBI-BLAST database to explore their taxonomy. Strain HF shared 100% sequence similarity to *P. lilacinum* and strain HQ displayed 99.82% sequence similarity to *P. chrysogenum* (Fig. 2). Therefore, strains HF and HQ were taxonomically identified as *P. lilacinum* and *P. chrysogenum*, respectively.

3.2. Optimization of carbon and nitrogen sources

High-molecular-weight organic pollutants, such as long-chain *n*-alkanes and polycyclic aromatics, cannot be easily biodegraded under natural conditions. In many cases, addition of exogenous organic matter as an auxiliary carbon and energy source can facilitate the biodegradation of organic pollutants. This process is known as co-metabolism (Medaura et al., 2021). To improve the biodegradation efficiency of recalcitrant hydrocarbons in oil sludge, the optimal carbon and nitrogen sources for *P. lilacinum* HF and *P. chrysogenum* HQ were selected through a single-factor experiment. Fungal growth with oil sludge as the sole

carbon source was inferior to that with oil sludge and a small amount of soluble starch as the carbon source (Fig. 1b).

The mean colony diameter of strain HF on the screening medium supplemented with mannitol and soluble starch was the largest (56.00 and 55.50 mm, respectively; Table 1). When glucose and sucrose were used as auxiliary carbon source, the mean colony diameter of strain HF was smaller (52.00 and 52.00 mm, respectively). When lactose was added as an auxiliary carbon source, the mean colony diameter of strain HF was only 45.67 mm. The largest mean colony diameter for strain HQ was observed with soluble starch and lactose as the supplementary carbon source (68.50 and 64.00 mm, respectively). When the auxiliary carbon source was changed to sucrose and mannitol, the mean colony diameter of strain HQ decreased to 54.50 and 52.33 mm, respectively. The mean colony diameter of strain HQ with glucose as an auxiliary carbon source was only 42.00 mm. The optimization results indicate that when growing on the medium containing oil sludge, the preferable carbon sources for *P. lilacinum* HF were mannitol and soluble starch, and those for *P. chrysogenum* HQ were soluble starch and lactose. Accordingly, soluble starch was selected as the optimal carbon source for both fungi in the biodegradation test.

Next, the optimal nitrogen sources for the two strains were screened (Table 1). Strain HF grew well on the screening medium with yeast extract and peptone as the auxiliary nitrogen source, forming large colonies with the mean growth diameters of 45.67 and 43.33 mm, respectively. Smaller colony diameters were observed with $(\text{NH}_4)_2\text{SO}_4$ and NH_4Cl as the auxiliary nitrogen source (mean diameter 40.33 and 39.00 mm, respectively). The growth diameter of colonies with urea as an auxiliary nitrogen source was the smallest at 30.33 mm. With regard to strain HQ, excellent growth was observed with NH_4Cl , $(\text{NH}_4)_2\text{SO}_4$, and yeast extract as auxiliary nitrogen sources, based on the large respective colony diameters of 60.67, 58.67, and 58.00 mm. A smaller colony diameter was observed with peptone (54.00 mm). The smallest colony diameter of 41.33 mm was observed with urea as the auxiliary nitrogen source. The optimization results suggest that *P. lilacinum* HF prefers organic nitrogen to mineral nitrogen. Yeast extract was a suitable nitrogen source for strain HF, whereas NH_4Cl , $(\text{NH}_4)_2\text{SO}_4$, and yeast extract were preferred by *P. chrysogenum* HQ for growth on the medium containing oil sludge. Therefore, soluble starch and yeast extract were selected as the optimal carbon and nitrogen sources for the two fungi in subsequent experiments.

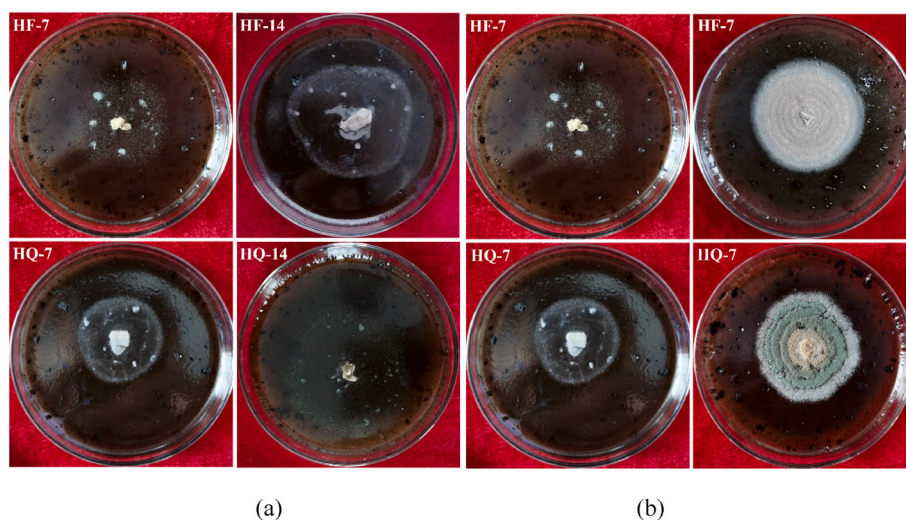


Fig. 1. Growth of two fungi HF and HQ on the mineral basal medium (a) containing oil sludge as the sole carbon source (7 days, left; 14 days, right) and (b) supplemented with 0.2% soluble starch as an additional carbon source (no starch, left; with starch, right; 7 days).

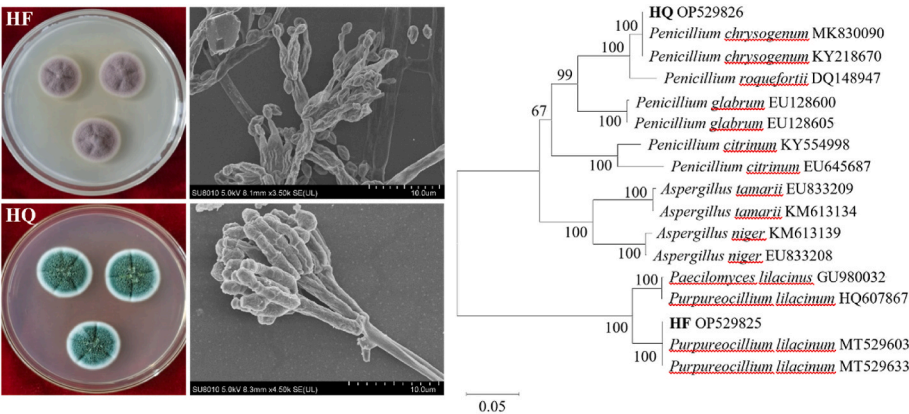


Fig. 2. Morphological features and phylogenetic relationship of two fungi HQ and HF.

Table 1

Colony diameter (mm) of two fungi HF and HQ grown on the oil sludge medium supplemented with different carbon and nitrogen sources for 14 days.

Isolate	Mannitol	Lactose	Sucrose	Glucose	Soluble starch
HF	56.00 ± 2.00 ^a	45.67 ± 1.15 ^d	51.50 ± 0.71 ^{bc}	52.00 ± 1.41 ^{bc}	55.50 ± 2.12 ^{ab}
HQ	52.33 ± 1.5 ^b	64.00 ± 3.0 ^a	54.50 ± 2.12 ^b	42.00 ± 2.83 ^c	68.50 ± 0.71 ^a
Isolate	Urea	NH ₄ Cl	(NH ₄) ₂ SO ₄	Yeast extract	Peptone
HF	30.33 ± 0.60 ^d	39.00 ± 0.00 ^c	40.33 ± 1.50 ^c	45.67 ± 0.60 ^a	43.33 ± 0.60 ^b
HQ	41.33 ± 3.50 ^c	60.67 ± 1.50 ^a	58.67 ± 0.60 ^a	58.00 ± 3.50 ^{ab}	54.00 ± 0.00 ^b

Values are presented as means ± standard deviation ($n = 3$). Different superscript letters within the same row indicate significant differences between groups ($P < 0.05$) according to Duncan's multiple range test.

3.3. Biodegradation capacity of selected fungi

The degradation rates of *P. lilacinum* HF and *P. chrysogenum* HQ for hydrocarbons in oil sludge were measured on the biodegradation medium supplemented with 0.2% soluble starch and yeast extract for 30 days. Supplementary Table S1, the TPH degradation rates of light oil were 33.81% (HF) and 24.01% (HQ), which were more than two times higher than those of oil sludge at 14.04% (HF) and 10.06% (HQ). Strain HF showed significantly higher biodegradation capacity for hydrocarbons in both light oil and oil sludge compared with strain HQ ($P < 0.05$).

To reveal the biodegradation capacity of the two fungi for various hydrocarbon components, GC-MS was conducted to analyze the original and residual oil/petroleum hydrocarbons extracted from the oil sludge samples (Fig. 3a). In the blank control, the relative abundances of different hydrocarbon components in light oil ranked in descending order of alkenes (32.95%) > iso-alkanes (24.90%) > cycloalkanes (23.27%) > aromatics (11.46%) > n-alkanes (7.41%). In the HF treatment, the degradation rates of all hydrocarbon components were

relatively high. In particular, *n*-alkanes were completely degraded (100%), followed by iso-alkanes (43.52%), alkenes (40.45%), and cycloalkanes (33.42%). The degradation rate of aromatics was the lowest at 28.38%. In the HQ treatment, only *n*-alkanes were degraded in light oil (52.07%), along with production of other hydrocarbon components. The relative abundance of aromatics increased the most (69.72%), and those of iso-alkanes, cycloalkanes, and alkenes also increased substantially (37.20%, 25.97%, and 24.38%, respectively).

The hydrocarbon components of oil sludge in the control group were dominated by *n*-alkanes (72.02%), followed in descending order by iso-alkanes (11.89%) > aromatics (7.98%) > alkenes (6.55%) > cycloalkanes (0.56%; Fig. 3b). In the treatment with strain HF, the degradation rate of cycloalkanes reached 100%, followed by those of *n*-alkanes (21.20%), aromatics (15.09%), and iso-alkanes (14.53%). The lowest rate was observed for alkenes (9.16%). Strain HQ was only able to biodegrade *n*-alkanes (1.33%), accompanied by production of other hydrocarbon components analyzed. The largest increase was observed for cycloalkanes (82.59%), and the relative abundances of iso-alkanes,

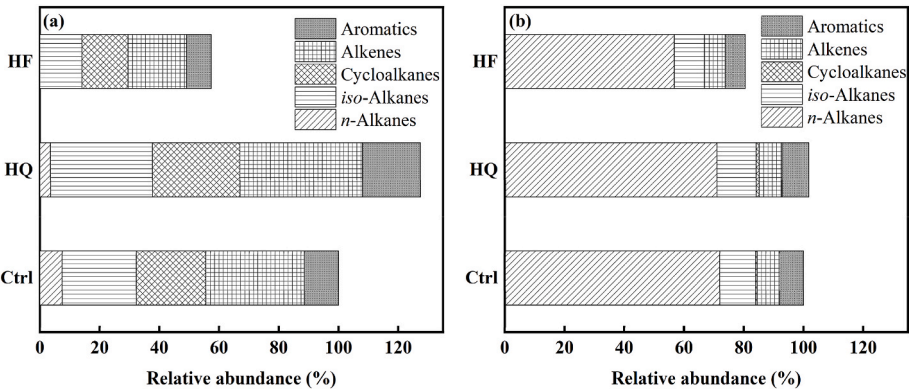


Fig. 3. Changes in the relative abundances of hydrocarbon components in (a) light oil and (b) oil sludge after fungal treatment based on GC-MS analysis.

aromatics, and alkenes increased less prominently (10.23%, 11.89%, and 1.59%, respectively). In the presence of oil sludge as a carbon source, both *P. lilacinum* HF and *P. chrysogenum* HQ grew vigorously (Fig. 1b) with moderate degradation of hydrocarbon components. These results were encouraging, but further analyses were still necessary to verify the potential of the two fungi for bioremediation of oil sludge-contaminated soils.

3.4. Removal and biodegradation of hydrocarbons in oil sludge

To evaluate the potential of *P. lilacinum* HF and *P. chrysogenum* HQ for bioremediation of oil sludge-contaminated soils, laboratory simulation experiments were performed. Both strains removed a proportion of hydrocarbons from modified oil sludge after 30 days of treatment. The removal efficiency of strain HF for hydrocarbons doubled that of strain HQ (15.31% vs. 7.62%, $P < 0.05$; Fig. 4).

The *n*-alkane fractions in oil sludge after biodegradation were analyzed by GC-MS (Table 2). The presence of strain HF led to variable decrease of all C_{9-26} *n*-alkanes in oil sludge with the mean degradation rate of 77.68%. When treated with strain HQ, C_{9-17} *n*-alkanes decreased considerably with the mean degradation rate of 31.77%, accompanied by an increase of C_{19-26} *n*-alkanes with the mean production rate of 5.58%. In contrast with *P. chrysogenum* HQ, *P. lilacinum* HF degraded diverse hydrocarbons in the light, medium, and heavy fractions (C_9 – C_{26}) of oil sludge.

4. Discussion

Microbial degradation is an emerging strategy for bioremediation of oil sludge-contaminated soils, and hydrocarbon biodegradation ensures maximum removal of TPH from oil sludge. Various fungi from genera that include *Aspergillus*, *Penicillium*, *Fusarium*, *Trichodema*, *Paecilomyces*, and *Talaromyces* have been recognized as petroleum hydrocarbon degraders. Notably, *Aspergillus* spp. and *Penicillium* spp. as effective hydrocarbon degraders can degrade and/or mineralize hydrocarbons to less hazardous compounds (Aranda et al., 2017; Essabri et al., 2019). *P. chrysogenum* HQ isolated in the present study belongs to the genus *Penicillium* that contains known hydrocarbon degraders. It shows the capacity to biodegrade various hydrocarbons under laboratory conditions probably owing to well-developed enzymatic systems that allow

the biodegradation of petroleum hydrocarbons to small molecules (Dhar et al., 2014). Considering the other strain (HF) isolated in the present study, *P. lilacinum* is rarely described as an efficient hydrocarbon degrader. The capability of *P. lilacinum* HF to biodegrade hydrocarbons was highlighted by its outstanding growth on the screening medium containing oil sludge (60 g L^{-1}) as the sole carbon source. The results indicate that both *P. lilacinum* HF and *P. chrysogenum* HQ are able to use hydrocarbons in oil sludge as a source of nourishment.

The core of successful soil remediation is biodegradation and removal of contaminants. High-molecular-weight organic contaminants cannot be easily biodegraded under natural conditions. Co-metabolism with microorganism is an effective strategy to deal with refractory organic compounds in terms of economy and technology (Tian et al., 2022). The underlying mechanism is that the co-occurrence of an additional carbon and energy source enhances the activity of microbial enzymes and improves the degradation efficiency of non-growth substrates (Medaura et al., 2021). In the present study, soluble starch and yeast extract were selected as the co-metabolism substrates for both *P. lilacinum* HF and *P. chrysogenum* HQ in the biodegradation test. The two fungi could grow well with soluble starch as an auxiliary carbon source, similar to *Pleurotus ostreatus* JY-11, which is able to degrade crude oil and produce oil-degrading enzymes (Wang et al., 2021). When soluble starch is sterilized at high temperature, the glycosidic bonds in the sugar chain are cleaved, and the resulting small-molecular-weight carbohydrates may be more easily assimilated and utilized by fungal hyphae (He et al., 2003). Co-metabolism has been widely used as a biotechnology in the biodegradation of organic compounds, such as aromatics (Wang et al., 2016).

The degradation rates of TPH in oil sludge by *P. lilacinum* HF (14.04%) and *P. chrysogenum* HQ (10.06%) observed in the present study fell in the range of 7.92%–25.60% previously rereported for other bacteria, such as *Acinetobacter* sp., and fungi, such as *Alternaria* sp. (Dai et al., 2021; Romero-Hernández et al., 2021). As expected, the degradation rates of TPH in oil sludge were lower than those of light oil observed in the present study and previous reports. For example, up to 30.43% and 44.55% of TPH were degraded by *A. ustus* HM3 and *P. lilacinum* HM4 within 40 days (Benguenab and Chibani, 2021). Our results were also lower than previously reported results for conventional heavy oil, where 25.29% and 27.36% of heavy oil were degraded by two *Aspergillus* isolates (*A. terreus* and *A. nidulans*) after 15 days of treatment. Because the hydrocarbons extracted from our oil sludge sample was dominated by asphaltenes, its inherent recalcitrance might limit biodegradation. Earlier studies have shown that it is extremely difficult to degrade asphaltenes biologically (Gao et al., 2017). *A. fischeri*, isolated from an asphalt lake in Venezuela, was the first microorganism able to grow using purified asphaltenes as the sole carbon source (Uribe-Alvarez et al., 2011). Recently, Romero-Hernández et al. (2021) isolated an *Alternaria* sp. from deep-sea sediments of the Gulf of Mexico, which was able to grow with extra-heavy crude oil as the sole carbon source, degrading up to 25.6% of the total crude oil.

In the present study, *P. lilacinum* HF and *P. chrysogenum* HQ were isolated from heavy oil sludge. This allows both fungi to utilize oil sludge hydrocarbons and thrive under adverse conditions without being denatured, despite their degradation rates for TPH are not higher than previously reported results. Indeed, microbial strains retrieved from oil sludge are presumably better adapted to this particular environment, which implies their great potential for application in bioremediation of oil sludge-contaminated soils. The degradation rate of TPH can be improved by creating favorable conditions or providing humic matter to facilitate the desorption of hydrocarbons from oil sludge and thereby enhance fungal activity. Furthermore, it is worth noting that fungal consortia exhibit a higher ability of petroleum hydrocarbon degradation than compared with individual isolates, most likely a result of positive synergistic effects (Hamad et al., 2021). This opens up new possibilities for oil sludge treatment using a consortium of *P. lilacinum* HF and *P. chrysogenum* HQ, which takes advantage of different biodegradation

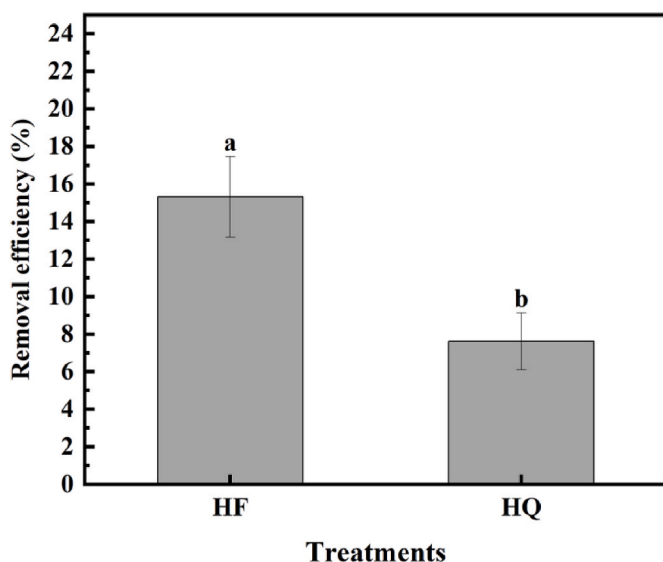


Fig. 4. Removal efficiency of hydrocarbons from modified oil sludge by two fungi HF and HQ. Error bars indicate standard deviation of the mean. Different letters above the error bars indicate significant differences between groups ($P < 0.05$) according to Duncan's multiple range test.

Table 2Biodegradation of *n*-alkane fractions in oil sludge by two fungi HF and HQ.

Retention time (min)	Molecular formula	Peak area Control	Peak area HF	Relative increase (%)	Peak area HQ	Relative increase (%)
969.9	C ₉ H ₂₀	2618161	1557844	-40.50%	2018101	-22.92%
1187.5	C ₁₀ H ₂₂	11240836	3381452	-69.92%	8646231	-23.08%
1390.7	C ₁₁ H ₂₄	34136517	7855271	-76.99%	16603278	-51.36%
1579.3	C ₁₂ H ₂₆	57966733	8873925	-84.69%	22843127	-60.59%
1754.7	C ₁₃ H ₂₈	100000000	24128429	-75.87%	56330307	-43.67%
1918.4	C ₁₄ H ₃₀	104000000	14858292	-85.71%	71150351	-31.59%
2072.3	C ₁₅ H ₃₂	112000000	22043430	-80.32%	95496142	-14.74%
2217.1	C ₁₆ H ₃₄	95610365	15443341	-83.85%	89650735	-6.23%
2353.9	C ₁₇ H ₃₆	85548179	16410609	-80.82%	84652701	-1.05%
2483.5	C ₁₈ H ₃₈	66928777	12460788	-81.38%	66932901	0.01%
2606.8	C ₁₉ H ₄₀	65525549	12233635	-81.33%	67078327	2.37%
2723.9	C ₂₀ H ₄₂	47716197	10547856	-77.89%	49522335	3.79%
2835.7	C ₂₁ H ₄₄	35752853	7092547	-80.16%	37598779	5.16%
2942.4	C ₂₂ H ₄₆	28964390	5497523	-81.02%	30662179	5.86%
3044.6	C ₂₃ H ₄₈	22740166	4781998	-78.97%	24234579	6.57%
3142.5	C ₂₄ H ₅₀	17759197	3758344	-78.84%	19519824	9.91%
3236.4	C ₂₅ H ₅₂	15000180	3714469	-75.24%	15926470	6.18%
3327.3	C ₂₆ H ₅₄	12592182	1930778	-84.67%	13191424	4.76%

mechanisms.

After 30 days of treatment, *P. lilacinum* HF and *P. chrysogenum* HQ induced remarkable changes in the hydrocarbon composition of oil sludge. Based on GC-MS analysis, we found that following the HF treatment, *n*-alkanes, *iso*-alkanes, cycloalkanes, and aromatics all decreased, whereas in the HQ treatment, only *n*-alkanes decreased, along with increased cycloalkanes, *iso*-alkanes, aromatics, and alkenes. Similarly, the results of hydrocarbon removal from oil sludge showed that HF utilized a wider range of *n*-alkanes, including aliphatics (C₉–C₂₆), whereas HQ exhibited different metabolic activity, leading to production of C_{19–26} *n*-alkanes. The biodegradation results of *P. chrysogenum* HQ are contradictory to the results of *P. lilacinum* HF (this study) and previously reported strains, which decreased the contents of all hydrocarbon fractions in oil sludge (Aranda et al., 2017; Hamad et al., 2021). However, all quantitative changes in the hydrocarbon components of oil sludge suggest the bioconversion of various fractions due to fungal degradation, which may positively influence the physical properties of oil sludge, and in turn facilitate hydrocarbon degradation (Mohamed et al., 2006; Dai et al., 2022). Different possible metabolic pathways for the fungal strains to degrade hydrocarbons in oil sludge are indicated in this study.

Bioremediation of oil sludge-contaminated soils necessitates the use functional microorganisms that can secrete specific metabolites and effectively degrade organic contaminants at highly polluted sites (Al-Hawash et al., 2018). In the present study, both *P. lilacinum* HF and *P. chrysogenum* HQ partly removed hydrocarbons from modified oil sludge within a 30-day period. Our result is similar to that of Zhou et al. (2019), who demonstrated fungal proliferation and hydrocarbon removal during biostimulation of oil sludge with high total petroleum. The authors described that TPH content in oil sludge decreased by 17.0% in the bioaugmentation treatment after a 13-week incubation. However, the removal efficiency of hydrocarbons from oil sludge observed in the present study is lower than previously reported results for light oil (Othman et al., 2020), where approximately 63% of TPH were removed within 16 days by *A. fumigatus* Shu2. Hydrocarbons in heavy oil sludge is difficult to degrade and remove mainly because of its high asphaltene content. Compared with previously reported hydrocarbon-degrading fungi, such as *A. ustus* and *P. lilacinum* (Ben-guenab and Chibani, 2021), *P. lilacinum* HF and *P. chrysogenum* HQ can utilize a wider range of hydrocarbon components in oil sludge, including *n*-alkanes (C₉–C₂₆), which signifies their superiority in degrading long chain *n*-alkanes. Biodegradation of long-chain *n*-alkanes can improve the bioavailability and biodegradability of recalcitrant hydrocarbons in oil sludge, which is of economic value.

5. Conclusions

Two hydrocarbon-degrading fungi, *P. lilacinum* HF and *P. chrysogenum* HQ, were isolated and screened from a heavy oil sludge sample. Both fungal strains grew well with oil sludge containing abundant hydrocarbons as the sole carbon source. When supplemented with soluble starch and yeast extract as the additional carbon and nitrogen sources, each strain was able to biodegrade a portion of hydrocarbons in oil sludge within 30 days. The composition of various hydrocarbon components in oil sludge changed after fungal treatment. HF could degrade a wide range of *n*-alkanes, *iso*-alkanes, cycloalkanes, alkenes, and aromatics in heavy oil at 280 °C, whereas HQ was able to degrade *n*-alkanes, accompanied by production of other hydrocarbon components. Petroleum hydrocarbons were effectively removed from oil sludge by fungal treatment mainly through the degradation of C_{9–26} *n*-alkanes (HF) or C_{9–16} *n*-alkanes (HQ). This study reports two candidate strains with great potential for bioremediation of oil sludge-contaminated soils. A consortium of *P. lilacinum* HF and *P. chrysogenum* HQ may be more efficient than single strains in removing hydrocarbons from oil sludge, as it takes advantage of different biodegradation mechanisms to facilitate the bioremediation process.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ibiod.2023.105566>.

References

- Al-Hawash, A.B., Alkooranee, J.T., Abbood, H.A., Zhang, J.L., Sun, J., Zhang, X.Y., Ma, F. Y., 2018. Isolation and characterization of two crude oil-degrading fungi strains from Rumaila oil field. Iraq. Biotechnol. Rep. 17, 104–109.
- Aranda, E., Godoy, P., Reina, R., Badia-Fabregat, M., Rosell, M., Marco-Urrea, E., García-Romera, I., 2017. Isolation of Ascomycota fungi with capability to transform PAHs: insights into the biodegradation mechanisms of *Penicillium oxalicum*. Int. Biodeterior. Biodegrad. 122, 141–150.
- Baoune, H., Aparicio, J.D., Pucci, G., El-Hadj-Khelil, A.O., Polti, M.A., 2019. Bioremediation of petroleum-contaminated soils using *Streptomyces* sp. Hlh1. J. Soil Sediment. 19, 2222–2230.
- Baron, N.C., de Souza Pollo, A., Rigobelo, E.C., 2020. *Purpureocillium lilacinum* and *Metarhizium marquandii* as plant growth-promoting fungi. PeerJ 8, e9005.
- Benguenab, A., Chibani, A., 2021. Biodegradation of petroleum hydrocarbons by filamentous fungi (*Aspergillus ustus* and *Purpureocillium lilacinum*) isolated from used engine oil contaminated soil. Acta Ecol. Sin. 41, 416–423.
- Bento, F.M., Camargo, F.A., Okeke, B.C., Frankenberger, W.T., 2005. Comparative bioremediation of soils contaminated with diesel oil by natural attenuation, biostimulation and bioaugmentation. Bioresour. Technol. 96, 1049–1055.
- Bovio, E., Gnani, G., Prigione, V., Spina, F., Denaro, R., Yakimov, M., Calogero, R., Crisafi, F., Varese, G.C., 2017. The culturable mycobiota of a Mediterranean marine site after the oil spill: isolation, identification and potential application in bioremediation. Sci. Total Environ. 576, 310–318.
- Chand, P., Dutta, S., Mukherji, S., 2022. Slurry phase biodegradation of heavy oily sludge and evidence of asphaltene biotransformation. J. Environ. Manag. 324, 116315.
- Cheng, L.J., Xue, Q.H., 2012. Microbiology Laboratory Technology, second ed. Science Press, Beijing.
- Dai, X.L., Lv, J., Guo, S.H., Wei, W.X., 2021. Heavy oil biodegradation by mixed bacterial consortium of biosurfactant-producing and heavy oil-degrading bacteria. Pol. J. Environ. Stud. 30, 71–80.
- Dai, X.L., Lv, J., Wei, W.X., Guo, S.H., 2022. Bioremediation of heavy oil contaminated intertidal zones by immobilized bacterial consortium. Process Saf. Environ. 158, 70–78.
- Deng, Z., Jiang, Y., Chen, K., Gao, F., Liu, X., 2020. Petroleum depletion property and microbial community shift after bioremediation using *Bacillus halotolerans* T-04 and *Bacillus cereus* 1-1. Front. Microbiol. 11 (1–8), 353.
- Dhar, K., Dutta, S., Anwar, M.N., 2014. Biodegradation of petroleum hydrocarbon by indigenous fungi isolated from ship breaking yards of Bangladesh. Int. Res. J. Biol. Sci. 3, 22–30.
- Dong, X.H., Liu, H.Q., Chen, Z.X., Wu, K.L., Lu, N., Zhang, Q.C., 2019. Enhanced oil recovery techniques for heavy oil and oilsands reservoirs after steam injection. Appl. Energy 239, 1190–1211.
- Ehiosun, K.I., Godin, S., Urios, L., Lobinski, R., Grimaud, R., 2022. Degradation of long-chain alkanes through biofilm formation by bacteria isolated from oil-polluted soil. Int. Biodeterior. Biodegrad. 175, 105508.
- Essabri, A.M.A., Aydinlik, N.P., Williams, N.E., 2019. Bioaugmentation and biostimulation of total petroleum hydrocarbon degradation in a petroleum-contaminated soil with fungi isolated from olive oil effluent. Water Air Soil Pollut. 230 (1–16), 76.
- Gallego-Velázquez, J., Cardona-Bustos, N.L., Restrepo-Betancur, F., 2014. Compatibility of the entomopathogenic fungus *Purpureocillium* sp. cepa UdeA0106 with biocontrollers fungi and plant protection products, used on chrysanthemum crops. Actual. Biol. 36, 172–186.
- Gao, H., Zhang, J.H., Lai, H.X., Xue, Q.H., 2017. Degradation of asphaltene by two *Pseudomonas aeruginosa* strains and their effects on physicochemical properties of crude oil. Int. Biodeterior. Biodegrad. 122, 12–22.
- Gnanasekaran, R., Iyyappan, J., Deepika, S., Mageshwari, R., Saranraj, S., Sathayanarayanan, V., 2019. Degradation of hydrocarbon by using isolated fungal species with laccase activity. J. Environ. Biol. 40, 796–801.
- Hamad, A.A., Moubasher, H.A., Moustafa, Y.M., Mohamed, N.H., Abd-el rhim, E.H., 2021. Petroleum hydrocarbon bioremediation using native fungal isolates and consortia. Sci. World J. 1–13, 2021.
- Haque, S., Srivastava, N., Pal, D.B., Alkhanani, M.F., Almalki, A.H., Areeshi, M.Y., Naidu, R., Gupta, V.K., 2022. Functional microbiome strategies for the bioremediation of petroleum-hydrocarbon and heavy metal contaminated soils: a review. Sci. Total Environ. 833, 155222.
- He, B.X., Xu, Y., Zhang, C.K., 2003. A study on assimilation of glucose and soluble starch of 8 mushrooms. J. Henan Institute Sci. Technol. 31, 28–30.
- Kong, H.Z., 2007. Annals of Fungi in China: *Penicillium* and Related Sexual Genera, first ed. Science Press, Beijing.
- Medaura, M.C., Guivernau, M., Ventas, X.M., Boldú, F.X.P., Viñas, M., 2021. Bioaugmentation of native fungi, an efficient strategy for the bioremediation of an aged industrially polluted soil with heavy hydrocarbons. Front. Microbiol. 12, 626436.
- Mikolash, A., Donath, M., Reinhard, A., Herzer, C., Zayadan, B., Urich, T., Schauer, F., 2019. Diversity and degradative capabilities of bacteria and fungi isolated from oil-contaminated and hydrocarbon-polluted soils in Kazakhstan. Appl. Microbiol. Biotechnol. 103, 7261–7274.
- Mohamed, M.E., Al-Dousary, M., Hamzah, R.Y., Fuchs, G., 2006. Isolation and characterization of indigenous thermophilic bacteria active in natural attenuation of bio-hazardous petrochemical pollutants. Int. Biodeterior. Biodegrad. 58, 213–223.
- Nwankwegu, A.S., Zhang, L., Xie, D., Onwosi, C.O., Muhammad, W.I., Odoh, C.K., Sam, K., Idenyi, J.N., 2022. Bioaugmentation as a green technology for hydrocarbon pollution remediation problems and prospects. J. Environ. Manag. 304, 114313.
- Othman, A.R., Ismail, N., Abdullah, S.R.S., Hasan, H.A., Kurniawan, S.B., Sharuddin, S.S.N., Ismail, N.I., 2020. Potential of indigenous biosurfactant-producing fungi from real crude oil sludge in total petroleum hydrocarbon degradation and its future research prospects. J. Environ. Chem. Eng. 10, 107621.
- Péguin, B., Cai, Q.H., Lee, K., Greer, C.W., 2022. Natural attenuation of oil in marine environments: a review. Mar. Pollut. Bull. 176, 113464.
- Pourfakhraei, E., Badraghi, J., Mamashli, F., Nazari, M., Saboury, A.A., 2018. Biodegradation of asphaltene and petroleum compounds by a highly potent *Daedaleopsis* sp. J. basic. Microb. 58, 609–622.
- Ramirez, D., Kowalczyk, R.M., Collins, C.D., 2019. Characterisation of oil sludges from different sources before treatment: high-field nuclear magnetic resonance (NMR) in the determination of oil and water content. J. Petrol. Sci. Eng. 174, 729–737.
- Rehman, R., Ali, M.I., Ali, N., Badshah, M., Iqbal, M., Jamal, A., Huang, Z.X., 2021. Crude oil biodegradation potential of biosurfactant-producing fungi *Pseudomonas aeruginosa* and *Meyerozyma* sp. J. Hazard Mater. 418, 126276.
- Romero-Hernández, L., Velez, P., Betanzo-Gutiérrez, I., Camacho-López, M.D., Vázquez-Duhalt, R., Riquelme, M., 2021. Extra-heavy crude oil degradation by *Alternaria* sp. isolated from deep-sea sediments of the Gulf of Mexico. Appl. Sci. 11, 6090.
- Saha, R.C., Reza, A., Hasan, M.S., Saha, P., 2019. A review-bioremediation of oil sludge contaminated soil. E3S Web Conf. 96, 01004.
- Sharuddin, S.S.N., Abdullah, S.R.S., Hasan, H.A., Othman, A.R., Ismail, N.I., 2021. Potential bifunctional rhizobacteria from crude oil sludge for hydrocarbon degradation and biosurfactant production. Process Saf. Environ. 155, 108–121.
- Su, H.F., Lin, J.F., Wang, Q.Y., 2021. A clean production process on oily sludge with a novel collaborative process via integrating multiple approaches. J. Clean. Prod. 322, 128983.
- Tian, Z.J., Li, G.W., Bai, M.X., Hou, X.L., Li, X.G., Zhao, C., Zhu, Q.H., Du, C.L., Li, M.T., Liu, W.J., Zhang, L.Y., 2022. Microbial mechanisms of refractory organics degradation in old landfill leachate by a combined process of UASB-A/O-USSB. Sci. Total Environ. 848, 157737.
- Uribe-Alvarez, C., Ayala, M., Perezgasga, L., Naranjo, L., Urbina, H., Vazquez-Duhalt, R., 2011. First evidence of mineralization of petroleum asphaltene by a strain of *Neosartorya fischeri*. Microb. Biotechnol. 4, 663–672.
- Wang, Q.Y., Zhang, G.Q., Chen, Q.J., Zhang, J., Li, X.F., Xu, S.Y., 2021. Screening and identifying highly efficient crude oil-degrading and degrading enzymes-producing fungi. Chin. J. Appl. Environ. Biol. 27, 1289–1296.
- Wang, S., Guo, S.H., Li, F.M., Yang, X.L., Teng, F., Wang, J.N., 2016. Effect of alternating bioremediation and electrokinetics on the remediation of *n*-hexadecane-contaminated soil. Sci. Rep. 6, 23833.
- Yang, G.C.C., Huang, S.C., Jen, Y.S., Tsai, P.S., 2016. Remediation of phthalates in river sediment by integrated enhanced bioremediation and electrokinetic process. Chemosphere 150, 576–585.
- Yanto, D.H.Y., Tachibana, S., 2014. Enhanced biodegradation of asphalt in the presence of Tween surfactants, Mn^{2+} and H_2O_2 by *Pestalotiopsis* sp. in liquid medium and soil. Chemosphere 103, 105–113.
- Zhang, J.H., Xue, Q.H., Gao, H., Lai, H.X., Wang, P., 2016. Bacterial degradation of crude oil using solid formulations of *Bacillus* strains isolated from oil-contaminated soil towards microbial enhanced oil recovery application. RSC Adv. 6, 5566–5574.
- Zhang, X.W., Zhang, H., Wang, H.X., Cao, Y.J., Zhang, L., 2022. A makeup remover-inspired chitosan-based emulsion for heavy oil removal in oily sludge treatment. Fuel 330, 125588.
- Zhao, Y.N., Yan, X.Y., Zhou, J.H., Li, R., Yang, S.A., Wang, B.F., Deng, R.N., 2019. Treatment of oily sludge by two-stage wet air oxidation. J. Energy Inst. 92, 1451–1457.
- Zhou, H.H., Huang, X.M., Bu, K.Y., Wen, F., Zhang, D.D., Zhang, C.F., 2019. Fungal proliferation and hydrocarbon removal during biostimulation of oily sludge with high total petroleum hydrocarbon. Environ. Sci. Pollut. Res. 26, 33192–33201.